A Machine Learning Investigation of Non-Cognitive Influences on Filipino Students' Math Performance in PISA 2022

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Introduction

The Philippines has first participated in the Program for International Student Assessment (PISA) last 2018 which evaluates outcomes of educational systems on learner acquired skills and competencies in Literacy, Science and Mathematics as well as non-cognitive factors like home and family characteristics, school environment, learner beliefs, attitudes and experiences among others (PISA, 2024; Govorova, Benítez, & Muñiz, 2020). This provided baseline information on the current educational landscape of the country. Last PISA 2022, the Philippines ranked 78 out of the 81 countries in Mathematics proficiency. Also, only 44% of the Filipino students who participated in PISA 2022 have reached the Level 2 and above proficiency or the basic proficiency in Mathematics.

The results have sparked greater interest and initiatives on the part of the government and other stakeholders to put forward agenda reforms in curriculum and likewise in teaching and learning delivery aspects of learning areas including mathematics.

The core of student achievement is the primary goal of the schools has been previously the "end all be all" of educational outcomes and research (Moore, 2019). However, much attention is equally given to non-cognitive factors that come to play a crucial role in student achievement and academic performance (Bernardo et al., 2022).

Using Machine learning modeling, the study aims to identify non-cognitive student factors that have contributed to the low performance of sampled 15-year old Filipino learners in the PISA 2022.

Methodology

The methodology of this study is patterned and adapted from the scholarly work of Dr. Allan B.I. Bernardo in his research article *"Bernardo, A. B., Cordel, M. O., Calleja, M. O., Teves, J. M., Yap, S. A., & Chua, U. C. (2023). Profiling low-proficiency science students in the Philippines using machine learning. Humanities and Social Sciences Communications, 10(1). doi:10.1057/s41599-023-01705-y"*

Data source

The dataset used in this analysis was from PISA 2022 data capturing only the Philippine dataset. This is a public domain that can be accessed at https://www.oecd.org/en/data/datasets/pisa-2022-database.html. For this study, only the Students' Questionnaires were considered. The dataset contains the feature values that were mostly processed using the GPCM or the Generalized Partial Credit Model. GPCM were used to establish the scale indices from the students' questionnaire. The questionnaires consist of items related to the student's background and school environment (*Pisa 2022 technical report* 2024).

According to the PISA 2022 Technical Report, the target population is composed of 15 year old students who are in grades 7 and higher. Also, the students must be enrolled full-time in educational institutions, under vocational training programmes, and those attending foreign schools. Not included in the sample are the students who are home-schooled. The PISA assessment used a two-stage stratified sampling design. First, schools with 15-year-old students (or the potential to have them) were selected from a national list of eligible schools, with larger schools having a higher chance of being chosen. Then, students were selected within those schools to participate in the assessment (*Pisa 2022 technical report* 2024).

Data Preparation

The original dataset was filtered based only on the derived variables since these already capture the individual questions that were grouped for certain vital non-cognitive measures.

The initial total number of considered features is 42. After removing the features with 100% missing values or no response, the final number of features is now 28. For the number of students, the initial count is 7,193. After removing the rows with more than 50% missing values, the final number of students is now 7,099 (Bernardo, Cordel, Calleja, Teves, Yap, & Chua, 2023). The remaining missing values were imputed using the k-nearest neighbor algorithm.

It was defined that proficiency Level 2 is the minimum requirement for high performing students (OECD, 2023). For this reason, the basis for the boundary value that was set for high and low performing students is the *PV1MATH* score of 358. Hence, the dependent variable was discretized into two classifications: "0" for high performing students or those with *PV1MATH* score of higher 358 otherwise the classification is "1" for low performing students.

Before processing	**Poor Performance (Level 1 and below)**	**Better performance (higher than Level 1)**	**Total**
Training Data	3189	2490	5679
Test Data	795	625	1420
Total	3984	3115	7099
After processing	**Poor Performance (Level 1 and below)**	**Better performance (higher than Level 1)**	**Total**
Training Data	2825	2828	5653
Test Data	795	625	1420
Total	3620	3453	7073

Table 1. Number of students split into training and test data before and after pre-processing to remove bias

The samples were split into a training set composed of 80% of the dataset and 20% for the test set.

As can be seen from Table 1, the number of samples for training set for the low performing students is higher than the high performing students. The dataset results in class imbalance and this can lead to biased behavior of the model training. To mitigate the bias, we performed undersampling of the majority class using the Synthetic Minority Oversampling Technique or SMOTE algorithm. For the minority class, the algorithm of Tomek Links was used which basically removes the noise, cleans the decision boundary and remove ambiguous or borderline instances (Bernardo, Cordel, Calleja, Teves, Yap, & Chua, 2023). After applying SMOTE and Tomek Links, the final number of training dataset is shown in Table 1.

Features

The Table 2 shows the considered features and description from the PISA 2022 dataset.

Feature Name	**Description**
Dependent Variable	
PV1MATH	Plausible values 1 for student's performance in Mathematics
Independent Variables	
ST004D01T	Gender
BULLIED	Exposure to bullying
BELONG	Sense of belonging
TEACHSUP	Teacher support in mathematics
ICTRES	Information and Communication Technology resources
ST016Q01NA	Life satisfaction
ST347Q01JA	Duration of school closures because of COVID-19
PROBSELF	Problems with self-directed learning
FEELSAFE	Feeling safe at school
FAMSUP	Family support
FEELLAH	Experience with learning at home
DISCLIM	Disciplinary climate in mathematics lessons

TARDYSD	Lateness
SKIPPING	Student truancy
MISSSC	Long-term student absenteeism at any education level
CREATOP	Creativity and openness to intellect
CREATEFF	Creative thinking self-efficacy
CURIOAGR	Curiosity
HOMEPOS	Home possessions
GROSAGR	Growth mindset
LANGN	Language spoken at home
IMMIG	Immigrant background
SISCO	Clear idea about future job
BSMJ	Expected occupation (OCOD3) and Expected occupation status
EXPECEDU	Highest expected educational level or Expected end of education (New for 2022)
HISEI	Highest parental occupational status
HISCED	Highest level of education of parents

Table 2. Features with desription

Exploratory Data Analysis

For the exploratory data analysis, the data visualizations that will be used are the bar chart to show the distribution of the students' scores in the Mathematics proficiency and the correlation matrix to show the direction of relationships of the features with the *PV1MATH* score.

Machine learning (ML) Model

Machine Learning algorithms are used to predict outcomes or the dependent variable given with several features or the independent variables. For this study, binary classification ML algorithms are considered since two outcomes are possible for each student, that is, low or high performing. ML algorithms are excellent in capturing the complex relationship between the features.

The ML algorithms that were considered in this study are Random forest, Logistic regression, XG Boost, SVM, and Decision trees. The description of each ML model and the Hyperparameters are shown in Table 3.

Model	Description	Hyperparameters
Random forest	An ensemble type that prevents overfitting by considering random individual trees and determining the majority voting.	n_estimators, min_samples_split,, min_samples_leaf, max_depth, bootstrap
Logistic regression	Utilizes the logistic or sigmoid function that predicts the outcome of either "0" or "1".	C, penalty, solver
XGBoost	A more efficient version of the Gradient Boosting because this performs parallel training of weak models.	C, degree, gamma, kernel
SVM	An algorithm that finds a hyperplane (or decision boundary) that best separates data into different classes.	criterion, max_depth, max_features:,min_samples_leaf,min_samples_split, splittert
Decision Trees	A simple algorithm that uses subsets of the dataset.	n_estimators, min_samples_split:, min_samples_leaf,max_depth, bootstrap

Table 3. Machine learning models with description and hyperparameters

Hyperparameter tuning was utilized to determine the optimal parameters that will give the highest accuracy score. For this study, the Randomized Search Cross Validation (*RandomizedSearchCV*) was used to optimize the hyperparameters. The *RandomizedSearchCV* was used because it is more efficient in terms of computation since it randomly selects combinations of predefined hyperparameters but still having a higher chance of providing the optimal set of parameters.

Figure 1 shows the process flow of training and testing of the ML model and subjecting it to model evaluation. The dataset will be split into 80% training data and 20% testing data. Then the training data will be used to train the ML model. The trained model will be used to test data then finally the model will be evaluated in terms of accuracy.

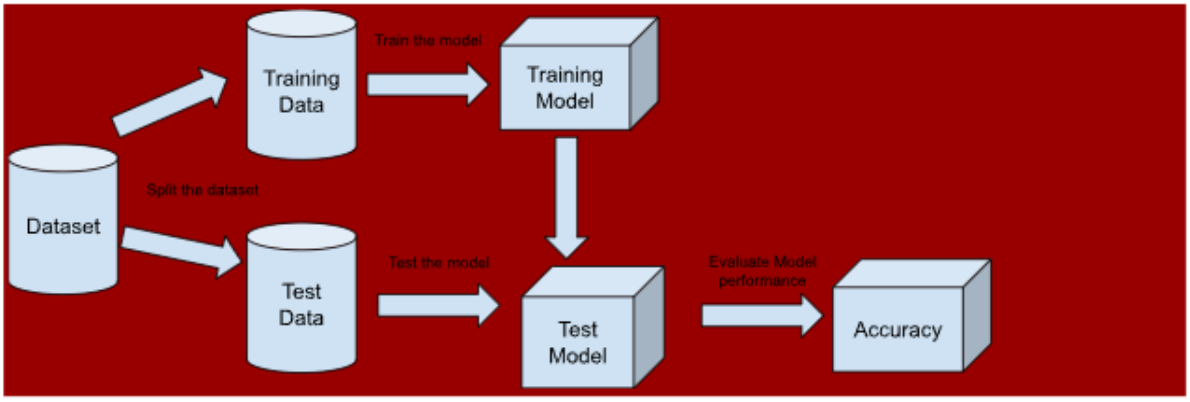


Fig 1. Flow process for the training, testing, evaluating the ML model

SHAP Value Analysis

SHapley Additive exPlanations is a model explainability method that is derived from cooperative game theory. This will serve as an aid in interpretation of the ML model's predictions. In this method, each feature will have assigned SHAP values that determine the feature's contribution to the prediction (Gnanasekaran, 2020).

SHAP values can be positive or negative. For binary classification, positive SHAP values indicate that the feature tends to drive the prediction toward the positive class while negative SHAP values drive the prediction toward the negative class. For this study, the positive class is the low performing category and the negative class is the high performing.

Data Results and Analysis

Exploratory Data Analysis

The bar chart in Figure 2 shows the distribution of students per proficiency level in Mathematics. Level 1 contains the most number of student. As the level increases, the composition of students decreases which indicates that 15 yr. old Filipino students are far from the highest level of proficiency.

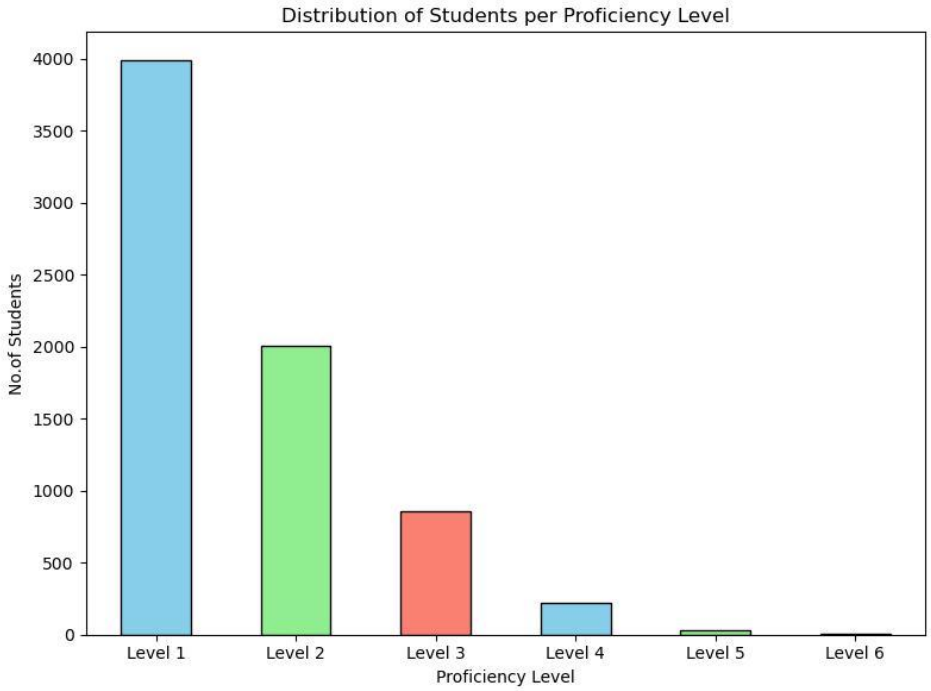


Fig 2. Distribution bar chart of students with Mathematics level proficiency

The correlation matrix of features is shown in Figure 3. The darker color scale indicates more negative correlation thus inverse relationship, while lighter color scale corresponds to more positive correlation or direct relationship. The features that are inversely correlated with the *PV1MATH* variable are the *MISSSC* and *BULLIED*. On the other hand, the features that are directly correlated with *PV1MATH* are *HISEI* and *CURIOAGR*. MISSSC pertains to long term absenteeism, specifically more than three consecutive months of absenteeism. This means that the longer the duration of absenteeism, the lower the *PV1MATH* score will be, thus being a low performer. Also, students who revealed that they were exposed to bullying have the tendency to have lower *PV1MATH* score. *HISEI* and *CURIOAGR* pertain to Highest parental occupational status and Curiosity respectively. Their relationship tells that the higher the occupational status of the parents, the higher the *PV1MATH* score will be. Also, the more open the student to learning the higher the *PV1MATH* score will be.

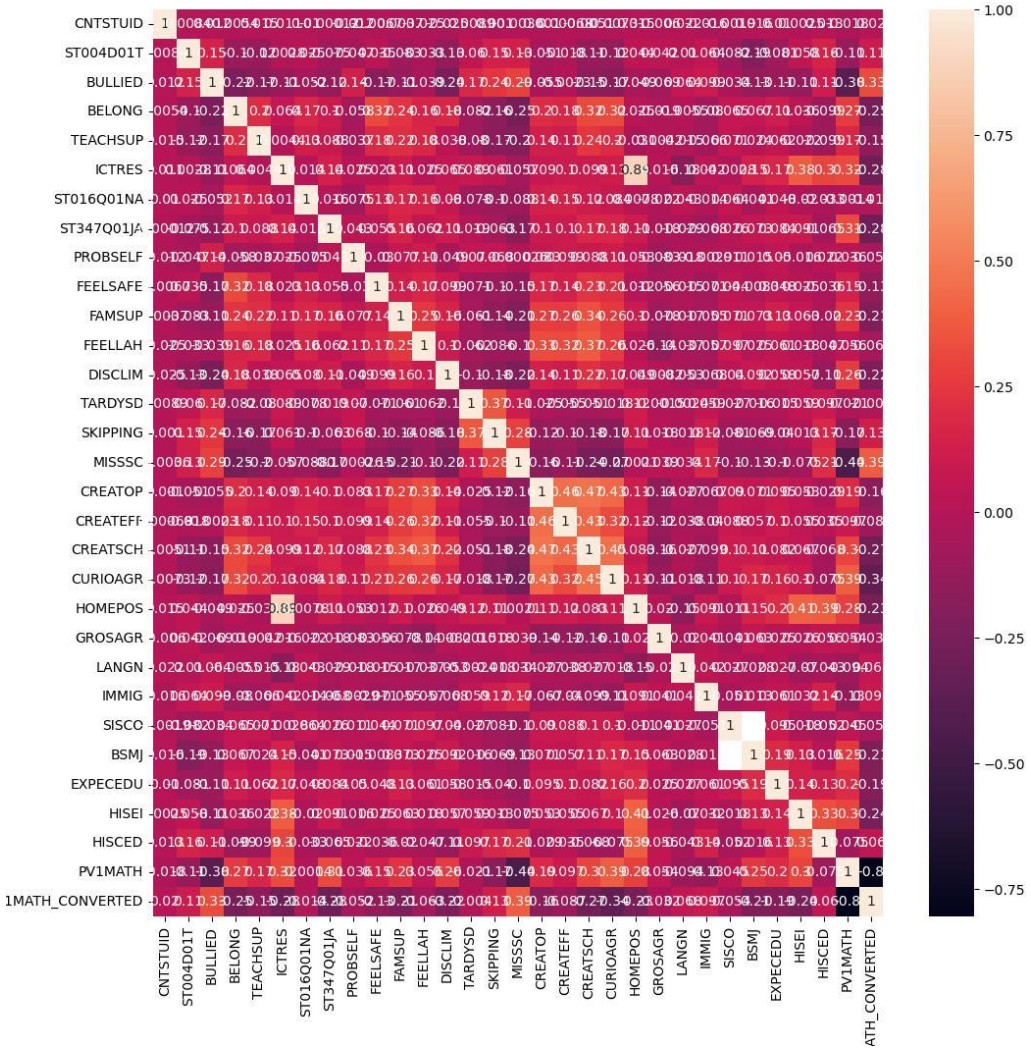


Fig 3. Correlation matrix of the features including the dependent variable PV1MATH

Machine learning algorithm selection

In terms of accuracy, the machine learning model that performs best is the XGBoost Classifier with accuracy of 0.779 and area under the curve of 0.85. The summary of the performance of all the models that were tested and evaluated is shown in Figure 4. The metrics of performance evaluation are shown in Table 4. The Confusion Matrix and ROC-AUC are also shown in Figures 5.1 and 5.2 respectively.

ML Model	Validation Performance				Hyperparameters Optimal Values
	Precision	Recall	F1-score	Accuracy	
Random forest	0.777	0.775	**0.775**	0.775	n_estimators: 200, min_samples_split: 2, min_samples_leaf: 1,max_depth:20, bootstrap: False
Logistic regression	0.77	0.771	0.771	0.771	C: 0.0028, penalty: L2, solver: lbfgs
XGBoost Classifier	**0.778**	**0.779**	0.773	**0.779**	subsample: 0.6, n_estimators: 500, min_child_weight: 4, max_depth:20, learning_rate:0.01,colsample_bytree :0.6
SVM	0.33	0.57	0.42	0.57	C: 4.210, degree: 4, gamma: 0.2886, kernel: rbf
Decision tree	0.679	0.665	0.667	0.665	criterion: gini, max_depth: 10, max_features: None,min_samples_leaf: 16,min_samples_split: 2, splitter: best

Table 4. Machine learning models with performance metrics

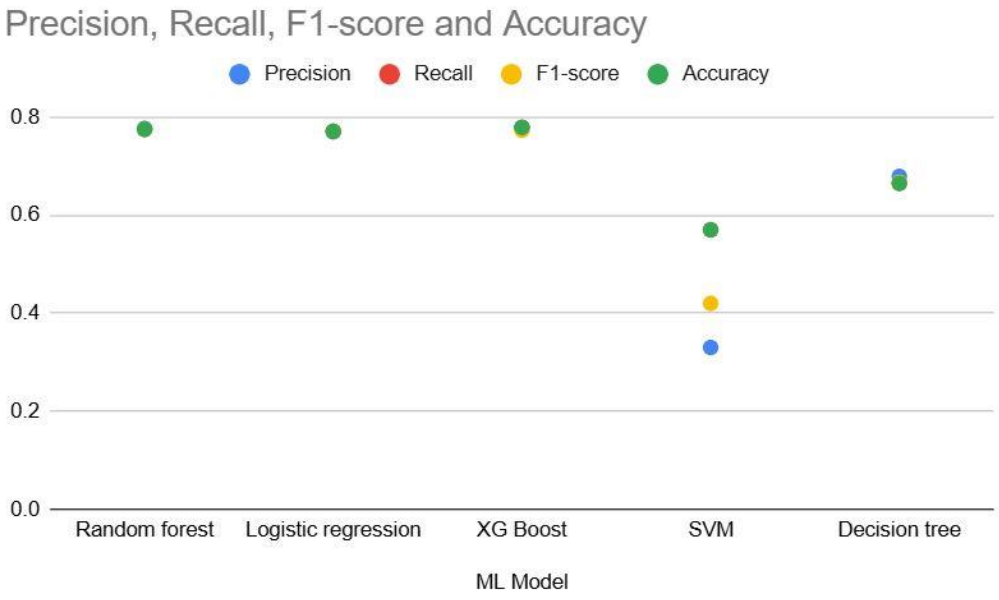


Fig 4. Summary of performance metrics using the Randomized Search Cross Validation

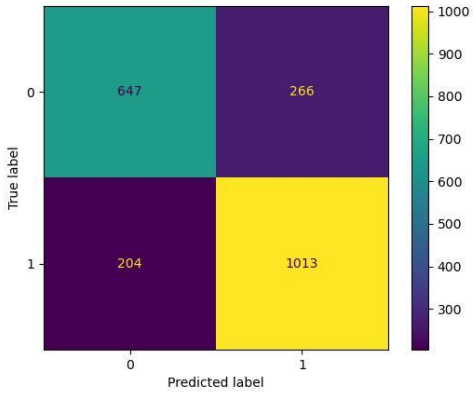


Fig. 5.1. The Confusion Matrix (XGBoost) with accuracy of 0.779

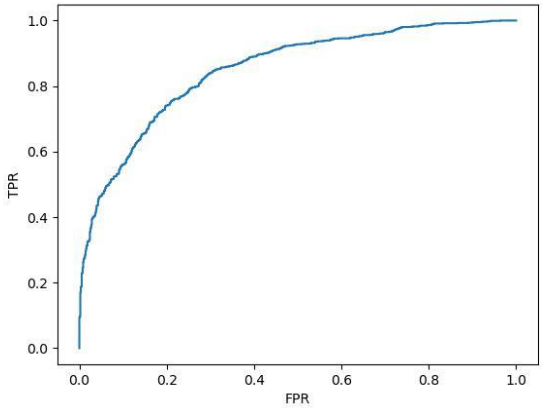


Fig. 5.2. ROC-AUC curve with 0.85 area under the curve

Feature importance using SHAP

The SHAP values for the features are shown in Figure 6. SHAP values are helpful for the model explainability. Only the ten features with highest SHAP values will be considered. For Figure 7, the summary plot of SHAP values is shown. It can be noticed that the dots correspond to the feature's value with color scale at the right most.

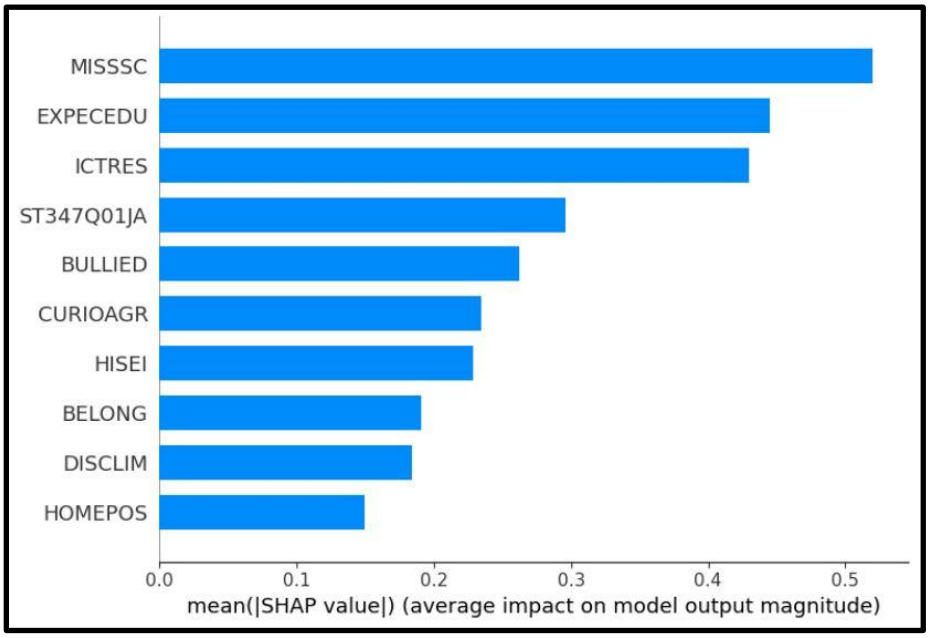


Fig 6. SHAP values for the top 10 features

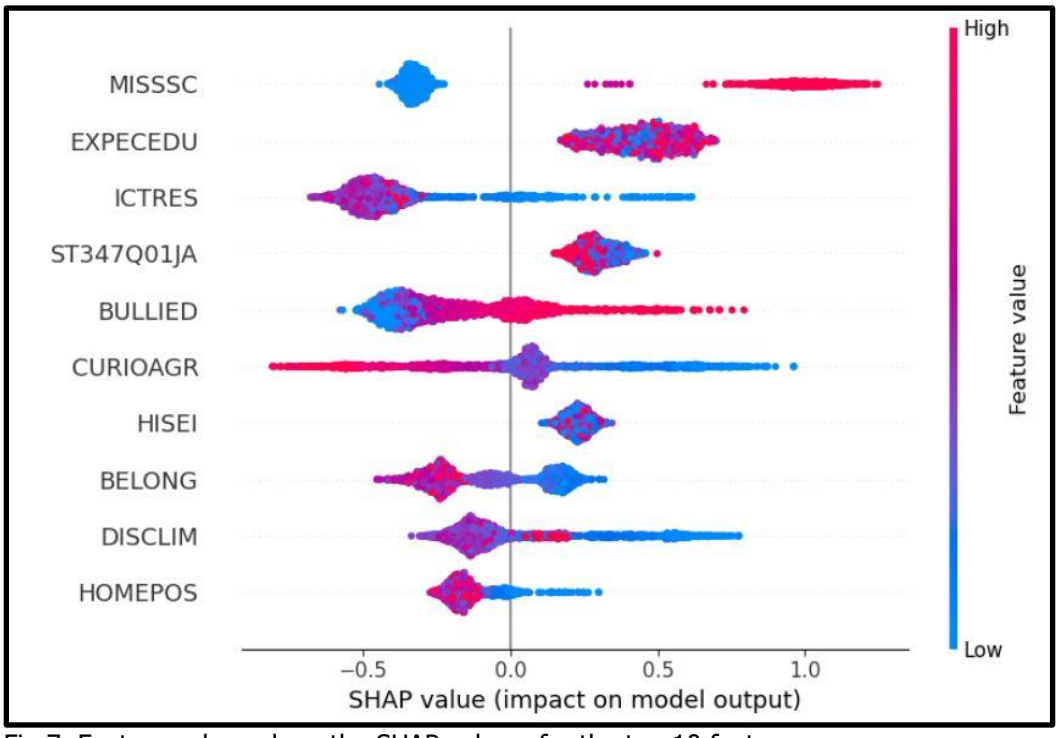


Fig 7. Feature values along the SHAP values for the top 10 features

MISSSC

The MISSSC feature refers to Long-term student absenteeism from primary to upper secondary school. This takes the value of "0" or "1", the latter indicating an absenteeism of more than three months in a row. From Figure 7, higher values of this feature correspond to positive SHAP values. This means that having more than 3 consecutive months of absenteeism is directly proportional to being low performing in Mathematics.

EXPECEDU

This feature describes the Highest expected educational level or Expected end of education which is a new feature for 2022 PISA. Higher values of this feature correspond to higher expected educational attainment of the student. From the SHAP plot, all values of this feature fall on the positive SHAP values. This may mean that whether they expect to attain the highest educational level, they will still tend to be low performing.

ICTRES

This feature refers to having access to Information and Communication Technology (ICT) resources. Examples of these resources include Educational Software or Apps, Internet access, digital devices at home such as television, desktop computers, laptops or notebooks, tablets, and cellphones. Positive values of this feature indicate having access to these ICT resources. As seen from Figure 7, lower values of this feature correspond to positive SHAP values. This means that students who have more access to ICT resources tend to be low performing in Mathematics.

ST347Q01JA
This feature corresponds to the question "In the last three years, was your school building ever closed for more than a week because of COVID-19?". The range of options are the following: 1-No, 2- Yes, up to 1 month, 3- Yes, more than 1 month and up to 3 months, 4-Yes, more than 3 months, 5-Yes, more than 6 months and up to 12 months, and 6- Yes, more than 12 months. Lower values of this feature correspond to short closure of school buildings during the pandemic. From the SHAP directionality plot in Figure 7, all values fall on the positive SHAP values. This reveals that regardless if the school building closure duration is shorter or longer, the student will be low performing in Mathematics.

BULLIED
The BULLIED feature refers to the exposure of students to bullying. The questions under this feature can be categorized to relational bullying, extortion bullying, verbal bullying, and physical bullying. The positive values of this feature mean that the student is more exposed to bullying than the average of the OECD. From Figure 7, the higher values correspond to positive SHAP values. This means that students who identify to be exposed to bullying tend to be low performing in Mathematics.

CURIOAGR
The CURIOAGR feature pertains to how much the student identifies to be curious or open to learning new things. The sample questions under this feature are "I am curious about many different things.", "I like to ask questions.", "I like to know how things work.", "I spend time to find more information about things that interest me." Positive values of this feature mean that the student is more curious than the average OECD. From Figure 7, lower values of this feature correspond to positive SHAP values. This means that students who are less curious tend to be low performing in Mathematics.

HISEI
This feature describes the Highest parental occupational status. Higher scores on this feature indicate higher levels of a parent's occupational status. From the SHAP plot, all values of this feature lie on the positive SHAP values. This means that regardless of the occupation attainment of the parent, the student tends to be low performing in Mathematics.

BELONG
This feature describes the Sense of Belonging of the student. Some of the statements in this feature are "I feel like an outsider (or left out of things) at school. ", "I feel awkward and out of place in my school.", and "I feel lonely at school." Positive values of this feature indicate a higher sense of belonging of the student. From Figure 7, lower values of this feature correspond to positive SHAP values which means that the students who identify to have lower sense of belonging tend to be low performing in Mathematics.

DISCLIM
This feature describes the Disciplinary climate in mathematics. Examples of the statements under this feature are "Students do not listen to what the teacher said.", "There is noise and disorder.", "The teacher has to wait a long time for students to quiet down.","Students cannot work well.", and "Students get distracted by using digital resources". These statements were described from 1 to 4, in which 1 corresponds to "Every Lesson", 2 as "Most lessons", 3 as "Some lessons", and 4 as "Never or almost never". Positive values of this feature correspond to a more conducive

learning environment. From the Figure 7, lower values of this feature fall on the positive range of SHAP values which pulls the result to be low performing.

HOMEPOS
This describes the wealth of the student's family or home possessions. Positive values of this feature mean that a student's family may have more possessions than the average OECD. From Figure 7, the lower values fall on the positive values which means that students with less wealth tend to be low performing in Mathematics.

Discussion

Student behavior and Learner Attitudes

Learner performance on mathematics achievement is critically affected by levels of learner absenteeism. Studies show consistency in the negative association of absenteeism and mathematics performance (Gottfried, 2015; Klein, Sosu, & Dare, 2022). The results of the study reveal the prevalence of absenteeism among Filipino learners, which may be caused by a myriad of factors like poverty, sickness or even lack of motivation to school. In the study of (Klein, Sosu, & Dare, 2022), absenteeism is tantamount to direct loss of learning, as students do not have access to instruction time which may lead to learning gaps. Absences when accrued, create cumulative disadvantages to learners resulting in further disengagement and widening of learning gaps and continued low academic performance.

Closure of Schools due to Covid, Learner Experiences and school environment

The effect of Covid-19 pandemic has transcended even in the delivery of basic education services. The study results provide that regardless of whether the teaching and learning happens in school or at home, students are bound to have low scores in math achievement. This shows that the available learning modality for students at the time is not effective for mathematics instruction.

The learning and social climate in schools appears to be a perennial concern among Filipino learners. The study results suggest that high incidences of bullying, low feelings of belongingness and a non-conducive learning environment due to a negative disciplinary climate are associated with poor mathematics performance among Filipino learners.

Learners who experience bullying are at the risk of having low achievement scores, including mathematics (Cheema & Kitsantas, 2013). In the study of Graham in 2023, traditional and cyberbullying has a negative association with mathematics achievement scores among African learners. The negative consequences of bullying include the experience of emotional and physical distress which impedes learners from focusing and engaging in their studies. (Bokhove, Muijs, & Downey, 2022). Bullying as a negative experience tends to affect learners for a prolonged time which becomes evident in their academic performance, emotional distress, behavioral concerns and even in motivations to learning and schooling.

This provides that bullying experience permeates in the functioning of learners over time as it evokes the presence of psychological difficulties (Murphy, Leonard, Taylor, & Santos, 2022). Per PISA 2022 results in the Philippines, bullying remains to be a rampant concern in schools Some

43% of girls and 53% of boys reported being the victim of bullying acts at least a few times a month (OECD average: 20% of girls and 21% of boys).

Much of what creates learner experience in the classroom and the school in general stems from the overall school climate (Wang, Perry, Malpique, & Ide, 2023) including the chance of getting higher academic achievement scores. Studies would affirm that when students perceive higher levels of discipline climate their math scores likewise increase (Cheema & Kitsantas, 2013). On the other hand, when learners feel connected with his peers and teachers, it provides a sense of emotional and social support that promotes engagement of learners - which in turn has an impact on ones mathematics achievement. (Korpershoek, Canrinus, Fokkens-Bruinsma, & de Boer, 2019).

Therefore, this puts a premium on the role of teachers to foster a positive classroom climate where learners are free from the experience of bullying, belongingness is felt from teachers and classmates and there is presence of discipline and among peers.

Access to internet and ICT

Digitization in education has transformed the landscape of teaching and learning in schools. Access to information and communications technology (ICT) including the internet has become crucial in student achievement and teaching and learning effectiveness. However, contrasting findings provide that access to ICT resources and internet does not automatically translate to student achievement in schools (Hampton, Robertson, Fernandez, Shin, & Bauer, 2021). In the study of Tan & Hew in 2017, students with access to Information Technology resources in schools take advantage of its benefits which promotes higher mathematics scores.

However, when internet use is excessive and spans up to a period of six hours, learners tend to have low mathematics scores (Rozgonjuk, Täht, & Vassil, 2021). When learners are left unsupervised, and inappropriately use ICT resources, students may engage in non-academic activities such as gaming and social media, which may interfere in mathematics learning and retention ((Gómez-García, Hossein-Mohand, Trujillo-Torres, Hossein-Mohand, & Aznar-Díaz, 2020; Hampton, Robertson, Fernandez, Shin, & Bauer, 2021).

Conclusion and recommendation

This study identifies non-cognitive factors, such as learning experiences, school climate, student behavior, and family and parental influences, as predictors of low performance in mathematical literacy. The findings offer a snapshot of the country's current educational landscape, underscoring the need for a paradigm shift.

Sustainable interventions should focus on enhancing teacher capacity in classroom management, implementing policies and programs that promote school belonging and discipline, encouraging the responsible and effective use of ICT resources in education, and strengthening home-school partnerships. These areas are crucial for tailoring education planning and delivery to students' specific contexts and needs.

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